MILITARY SPECIFICATION

TESTING KIT, IMPREGNITE IN CLOTHING, ABC-M2

1. SCOPE

1.1 This specification covers one type of testing kit for use in qualitative determinations of impregnite in clothing.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Box, Fiberboard.
PPP-C-843	-	Cushioning Material, Cellulosic.

MILITARY

MIL-P-3938	-	Pallets, Material Handling, Hardwood, Stringer Construction, 4-Way (Partial).

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-147	-	Palletized and Containerized Unit Loads.

FSC 8831

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DRAWINGS

US ARMY MUNITIONS COMMAND

DL5-77-236 - Testing Kit, Impregnite in Clothing, ABC-M2.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing DL5-77-236 or on the subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the kit shall conform to the specifications and drawings listed on Drawing D15-77-236 and subsidiary drawings.

3.2 Manufacture and assembly. The kit shall be manufactured and assembled as specified on Drawing C5-77-236.

3.3 Vial filling. The glass vials shall be filled with one of the applicable solutions as specified in table 1.

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Table I. Vial Filling

Ingredients	Specification	Parts by weight: Jackets and trousers	Parts by weight: Gloves and socks	Test method
Rhodamine B base (greater than 88.5% purity)	National Aniline Division Allied Chem & Dye Corp or equal	*	*	4.3.4.1
N, N Dimethyl formamide (bp 150° - 152°C; refractive index at 20°C - 1.4303 + 0.0005)	Eastman White Label 5870 or equal	60 ± 5	60 ± 5	4.3.4.2
Chlorobenzene (bp 130° - 132°C refractive index at 20°C - 1.5244 + 0.0005)	Eastman White Label 70 or equal	30 ± 3	30 ± 3	4.3.4.2
Ethylene glycol monomethyl ether (bp 120° - 125°C refractive index at 20°C - 1.4023 + 0.0005)	Eastman Practical Grade P2361 or equal	10 ± 2	10 ± 2	4.3.4.2

* The amount of rhodamine B base used shall provide concentrations equivalent to 2.70 ± 0.13 gram (g) of the standard rhodamine B for jackets and trousers and 1.35 ± 0.07 g of the standard rhodamine B for gloves and socks.

3.3.1 Ampoules. Each ampoule shall be filled with 0.05 ± 0.01 g of 2:1 water: acid solution by weight using distilled water and ACS reagent grade hydrochloric acid when tested as specified in 4.3.4.3.

3.4 Preproduction. Prior to the start of regular production, a preproduction sample of testing kits shall be produced for examination and test in accordance with 4.2.

3.5 Workmanship. The kit shall be free from defects (broken or cracked vial or ampoule and torn or broken box) and foreign matter (grease, oil and dirt) which would affect its suitability for the intended use.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier

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may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence accpetable to the contracting officer that the requirements of 3.1, 3.2 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Preproduction.

4.2.1 Sample. A preproduction sample of 10 testing kits shall be manufactured using the same materials, equipment and processes as will be used during regular production.

4.2.2 Inspection. The sample kits shall be examined in accordance with the classification of defects (4.3.3.3) and tested as specified in 4.3.4.

4.2.3 Acceptance/rejection criteria. The preproduction sample shall comply with the examination and tests specified in 4.2.2 to be acceptable.

4.3 Regular production.

4.3.1 Lotting. A lot shall consist of the kits produced by one manufacturer, at one plant, from the same materials under essentially the same manufacturing conditions. However, no more than one lot of reagents shall be represented in any one lot of finished testing kits.

4.2.2 Sampling.

4.3.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.3.2.2 For test. Five testing kits shall be taken at random from each lot for the tests specified in 4.3.4.

4.3.3 Inspection.

4.3.3.1 For examination. The sample kits shall be examined in accordance with the classification of defects and MIL-STD-105.

4.3.3.2 For test. The sample kits shall be tested in accordance with 4.3.4. If any of the sample kits fail to pass any of the test specified, the lot represented by the sample shall be rejected.

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4.3.3.3 Classification of defects.

(a) Testing kit, impregnite in clothing, ABC-M2 (Dwg C5-77-236).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Component missing or incorrect
102	Tape not wrapped securely around cap and neck of vial
103	Separator missing or incorrect
104	Marking on box or bottles missing, incorrect or illegible
105	Workmanship (see 3.5)

4.3.4 Tests. Tests shall be conducted as follows:

4.3.4.1 Rhodamine B.

(a) Preparation of the standard. Using a Government furnished standard sample of rhodamine B (see 6.3) accurately prepare five solutions in the range of 1.25 to 3 g of dye per 100 g of solvent (solvent is a 60/30/10 parts by weight mixture of N, N-dimethyl formamide, chlorobenzene, and ethylene glycol monomethyl ether). Take 10 milliliter (ml) samples of each soltuion and dilute with a constant volume of solvent mixture. For this dilution use a 50/49/1 parts by volume of isopropanol, distilled, water, and 28-percent ammonium hydroxide solution (see Note 1 below). Run the sample in a spectrophotometer at a wave length of 553 mμ to determined absorbance. If necessitated by the type of instrument employed, the fluorescence deviations in the test should be minimized by the proper location of the sample in the instrument. Plot absorbance versus grams of dye per 100 g of solvent in the known solutions.

NOTE 1. A dilution of 1:10,000 is satisfactory for absorbance measurements on an instrument of comparable quality to a Beckman D. U. Spectrophotometer.

(b) Procedure. Take 10 ml samples of the solutions to be tested and dilute in the same manner as in (a) above (same dilution and diluent). Run the sample in the spectrophotometer at a wave length of 553 mμ to determine absorbance. Determine the grams of dye per 100 g of solvent in the unknown using the standard calibration curve.

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4.3.4.2 Solvent determination (see 6.4). Determine the quantity of the three solvents in the mixture using gas chromatography. Use a column of sufficient size and length and packing impregnated with material which will effectively separate the three solvents with respect to time of elution. Using a 60/30/10 parts by weight mixture of N, N-dimethyl formamide, chlorobenzene, ethylene glycol monomethyl ether, determine sample size required in the chromatographic instrument to give optimum readings on the recorder chart. Prepare standard curves of integrated area versus quantity for each of the three solvents (these areas are read directly from the chart). Using the optimum sample size (determined above) of the solution to be tested, run the sample through the chromatography apparatus. Determine the percent by weight of each component by reading the quantity of component from standard curves for the areas obtained in the test.

4.3.4.3 Ampoules. Place an ampoule in an Erlenmeyer flask and break ampoule. Add 25 ml water, phenolphthalein indicator, and titrate to end point with 0.01N sodium hydroxide (NaOH). Calculate the grams of 2:1 hydrochloric acid (HCl) as follows:

Grams of 2:1 HCl = ml NaOH x normality of NaOH x 0.30

5. PREPARTION FOR DELIVERY

5.1 Packing.

5.1.1 Level A. Sixty testing kits shall be packed and cushioned in either a wooden box conforming to class 2, style 4-1/2, type 2 box of PPP-B-621 or a fiberboard box conforming to style RSC, grade V3c, type 2 (average load) box of PPP-B-636. Cushioning material conforming to type II, class A or B, of PPP-C-843 shall be used to prevent movement of the kits within the box. The wooden box shall be closed in accordance with the applicable requirements specified in the appendix to PPP-B-621. The fiberboard box shall be closed in accordance with the requirements for closure by taping class weather resistant, style RSC boxes to be shipped as exterior containers, as specified in the appendix to PPP-B-636. The wooden box shall be reinforced by galvanized steel strapping in accordance with the reinforcing requirements in PPP-B-621; and the fiberboard box shall be reinforced by tape banding.

5.1.2 Level C. The testing kits shall be packed as specified in 5.1.1 except that the fiberboard box shall be type CF, class domestic, grade 175 of PPP-B-636.

5.2 Marking. In addition to any special marking required in the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

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5.3 Palletization. When specified in the contract or order (see 6.2) shipping containers of 5.1 shall be palletized on a 40 by 48 inch pallet of MIL-P-3938 in accordance with the applicable requirements of MIL-STD-147.

6. NOTES

6.1 Intended use. The kit covered by this specification is intended for use in qualitatively determining whether impregnated clothing containes the required amount of impregnite necessary for protection.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Level of packing required. If level A, the type box required.
(c) Palletization required.
(d) Preproduction.

(1) Time allowed for supplier submission of sample for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instruction when testing is performed by the Government.
(3) Time allowed for the Government to notify the supplier whether or not to proceed with production.

6.3 Standard samples. Standard samples for use in determining the rhodamine B content in accordance with 4.3.4.1 may be obtained from Commanding Officer, Edgewood Arsenal, ATTN: SMUEA-QAIC, Edgewood Arsenal, Maryland 21010.

6.4 Solvent determination. The following procedure has been successfully utilized to separate the solvents:

Method - Gas chromatography using Fisher-Gulf Partitioner Model 160

(1) Column: Copper tubing 1/4-inch diameter - 6 feet long.
(2) Column packing: Columnpak 30 to 60 mesh - distributed by Fisher Scientific Company.
(3) Substrate: (a) Dioctylphthalate, 35 g per 100 g of Columnpak.
(b) Lac 3R728 supplied by Cambridge Industries Company, 15 g per 100 g of Columnpak.

(4) The substrates were put on the Columnpak separately from acetone solutions. The excess acetone was boiled off on a steam bath. Three feet of the column were packed with dioctylphthalate substrate and three feet of the column were packed with Lac 3R728.

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(5) a. Flash evaporator temperature 170°C.
b. Oven temperature 120°C.
c. Flow rate of carrier gas <60 ml/sec at 20 lb pressure
d. Carrier gas helium
e. Sample size 0.015 ml
f. Sensitivity used 15.0
g. Time of elution, minutes
Chlorobenzene 6.0
Ethylene glycol monomethyl ether 12.5
Dimethylformamide 21.5

(6) The sample was applied to the chrmoatograph with a pipette sampler.

6.5 Caution. The solvents used in the manufacture of ABC-M2 testing kits are flammable and toxic. Caution should be exercised in the handling and mixing of the solvents.

6.6 International interest. Certain provisions of this specification are the subject of international standardization agreement (see Serial No. 19, Supplement B to Quadripartite Standardization List Sheet Report). When amendment, revision or cancellation of the specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

Custodian:
Army - MU

Preparing activity:
Army - MU (EA)

International interest (see section 6)

Project No. 6630-0120

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